ELECTRIC VEHICLE MARKET PERFORMANCE & INSIGHTS

Updated Project Report

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Project components: Tableau Public dashboards and story • Python/Flask companion app • Static website and GitHub repository

Updated against the project repository and the current HTML page reference supplied by the author.

1. Introduction

1.1 Project Overview

Electric Vehicle Market Performance & Insights analyses an electric-vehicle market dataset and presents the findings through Tableau visualizations, a guided Tableau Story, a Python Flask companion application, and project website files hosted in GitHub. The report follows the existing project documentation while updating the website/file references to include the repository page named **EV Market Performance.html**.

Dataset: Electric Vehicle Market Classification Dataset. The repository stores the dataset in two compressed files: **compressed_data.csv.gz** and **compressed_data-2.csv.gz**.

Dataset source: Kaggle — Electric Vehicle Market Classification Dataset

1.2 Purpose and Objectives

- Turn a large EV dataset into visuals that can be understood quickly.
- Compare manufacturers, battery technologies, market regions and vehicle categories.
- Explore customer ratings, driving range, battery capacity and CO■-savings fields where available.
- Publish the analysis and supporting project files so the work can be reviewed and shared.
- Document the data workflow, dashboard design, testing, limitations and future improvements.

1.3 Project Links

Resource	Link
GitHub repository	EV-Market-Performance-Insights
HTML page	EV Market Performance.html
GitHub Pages website	Project website
Tableau dashboard	Executive Dashboard
Tableau Story	Green Executive Refresh — Story 2

Note: the repository page is large, and GitHub's rendered view does not expose the full HTML source in its web preview. The report therefore records the page as a current repository artifact without inventing unverified interface features.

2. Ideation Phase

2.1 Problem Statement

The dataset contains many records and fields, making it difficult to identify market patterns by reading the raw table. A filterable visual presentation helps users compare key measures and explore the data more efficiently.

2.2 Empathy Map

Element	User perspective
Says	Which manufacturers and regions perform best?
Thinks	The raw data needs a clear summary.
Does	Scrolls through tables and compares values manually.
Feels	Overwhelmed by many columns and figures.
Pains	Manual comparison is slow and there are few visual cues.
Gains	A dashboard with KPIs, filters, charts and a guided story.

2.3 Brainstorming

- KPI cards for market measures such as sales, ratings, driving range, battery capacity and CO■ savings, subject to the fields and calculations in the workbook.
- Charts for manufacturer sales, battery type, region, vehicle category, customer rating and range.
- A manufacturing-country map and interactive filters to support exploration.
- A Tableau Story that presents the analysis in a guided sequence.
- A Flask companion app and static website files for sharing the project.

3. Requirement Analysis

3.1 Customer Journey

Awareness	Exploration	Analysis	Decision	Follow-up
Opens the website or dashboard	Uses available filters	Reads charts, map and story	Identifies patterns in manufacturers, regions and battery types	Shares the project link or report

3.2 Solution Requirements

Type	Requirement	Detail
Functional	Interactive dashboard	KPIs, charts, map and filters where configured
Functional	Data preparation	Read compressed CSV files; remove duplicate rows during processing; handle missing values
Functional	Story	Guided sequence explaining the analysis
Functional	Publishing	Publish Tableau deliverables and share repository/website links
Functional	Website	Include the updated EV Market Performance.html page in project documentation
Non-functional	Usability	Readable labels, titles, legends and tooltips
Non-functional	Performance	Keep interactions responsive and avoid unnecessary data processing

3.3 Data Flow

Kaggle dataset → compressed CSV files in GitHub → data preparation and analysis → Tableau workbook/dashboard/story and Flask data processing → public project links and website files → end user.

3.4 Technology Stack

Layer	Technology
Data	Electric Vehicle Market Classification Dataset; CSV and gzip
Visualization	Tableau Desktop and Tableau Public
Backend / processing	Python, Flask and pandas
Frontend	HTML, CSS and JavaScript
Hosting / version control	GitHub, GitHub Pages and Tableau Public

4. Project Design

4.1 Problem–Solution Fit

The project addresses the difficulty of interpreting a large EV dataset by bringing key measures, charts, filters and geographic views into a visual analysis. The Tableau deliverables support analysis, while the website and repository provide entry points to the work and its supporting files.

4.2 Proposed Solution

- Store the source data in compressed files without modifying the original source files.
- Use Tableau worksheets and dashboards to compare relevant EV market dimensions.
- Publish the workbook and Story through Tableau Public.
- Maintain the Flask companion app for local data processing and the static website files for browser-based access.
- Document the current HTML artifact as **EV Market Performance.html** in addition to the repository's other website/project files.

4.3 Solution Architecture

Layer	Role
Data layer	Compressed CSV dataset parts
Processing layer	Tableau extract/workbook and pandas processing in Flask
Presentation layer	Tableau dashboard and Story; static HTML website page; Flask web interface
Access layer	Browser links to GitHub, GitHub Pages and Tableau Public; Flask app runs locally unless deployed to a Python-capable host

4.4 Deployment Notes

GitHub Pages serves static assets; it does not run a Python Flask server. The Flask application must be run locally or deployed to a hosting service that supports Python. Tableau Public workbooks and their published data extracts are public, so only data suitable for public sharing should be used.

5. Project Planning & Scheduling

5.1 Project Plan

Task	Activity	Planned duration	Status in prior report
1	Dataset study, download and cleaning	4 days	Completed
2	Tableau worksheets and calculated fields	4 days	Completed
3	Dashboard design, filters and Story creation	4 days	Completed
4	Tableau Public publishing and troubleshooting	4 days	Completed
5	Flask app, GitHub upload and report	4 days	Completed

5.2 Problems Faced During Development

The earlier report describes difficulty choosing the correct Tableau publishing destination. The resolution was to publish through Tableau Public using a saved extract rather than relying on a live connection to local files. This allowed the dashboard and Story to be shared using public links.

5.3 Current Repository Documentation

The repository README describes the project as an interactive EV market analysis built with Tableau Public, a Python Flask app and a static website. It also documents the compressed data parts and explains that Flask is separate from the Tableau Public deliverable. The HTML file supplied for this update is linked in Section 1.3.

6. Functional and Performance Testing

Test	Description	Status / evidence
Filter test	Manufacturer and other configured filters update views	Previously reported as working; recheck after workbook edits
KPI check	Displayed KPI values agree with the data and definitions	Previously reported as working; verify after data/logic changes
Publishing test	Tableau dashboard and Story open from public links	Links recorded in this report
Repository link test	GitHub repository and HTML page are reachable	Repository and HTML page links included
Static website test	Website loads in a browser through GitHub Pages	Recheck the published page after the latest commit
Flask app test	Application starts and serves its page locally	Run locally to confirm; not independently executed for this report

Testing note: statuses that depend on running the current application are not represented as newly verified. Re-run these checks after the latest code changes.

7. Results

7.1 Reported Dashboard Measures

The previous report recorded the following dashboard KPI values. They are preserved here as previously reported values, not independently recalculated during this update.

Measure	Previously reported value
Manufacturers	200
EV records	17,980
Total sales	1.55B
Average customer rating	4.09 / 5
Average driving range	324.5 km
Average battery capacity	78.2 kWh

7.2 Project Outputs

- Tableau workbook containing worksheets, dashboards, filters and Story.
- Published Tableau dashboard and Story links.
- Flask application and compressed dataset parts in the repository.
- Static website files, including the referenced **EV Market Performance.html** page.
- Project report and README documentation.

7.3 Interpretation and Limitations

The visualizations provide a convenient way to compare values in the dataset. Sales and CO■-related totals should not automatically be interpreted as market-wide estimates without confirming the meaning, coverage and unit of each row. The dashboard figures should be revalidated if the source data, filters or calculations have changed.

8. Advantages and Disadvantages

Advantages	Disadvantages / limitations
• Interactive charts and filters support exploration. • KPIs and geographic views summarize several dimensions. • Tableau Public and GitHub make the work easy to share. • Repository files support review and reproducibility.	• Tableau Public exposes published workbook data. • Publishing and viewing require internet access. • Dataset totals may not represent the entire EV market. • GitHub Pages cannot execute the Flask backend. • Current code changes require regression testing.

9. Conclusion

The project brings an EV dataset into a visual analysis workflow using Tableau, Python/Flask and website files hosted with GitHub. The updated report records the current repository HTML page reference while retaining the existing dashboard, Story and data-processing context. The result is a shareable student project that demonstrates data preparation, visualization, publishing and technical documentation.

10. Future Scope

- Add forecasting for future EV sales, with assumptions and model performance documented.
- Explore machine-learning classification of EV market segments.
- Deploy the Flask app to a Python-capable host and document the deployment steps.
- Add further data sources, such as charging infrastructure and price trends, if reliable data is available.
- Add screenshots of the current HTML page and retest links, filters and charts after each significant update.

11. Appendix

11.1 Repository and Supporting Files

GitHub repository

The repository README identifies the Tableau workbook, Flask application, compressed dataset files and static website as project components. The updated HTML page referenced for this report is available at:

EV Market Performance.html — GitHub

11.2 Dataset Link

Kaggle dataset

11.3 Run the Flask Application

The repository README recommends Python 3.11 or later, a virtual environment, and the Flask and pandas packages. From the project directory, activate the environment, install dependencies, run the application, and open http://127.0.0.1:5000. Keep the terminal open while using the local app. Exact startup behavior should be confirmed against the current application file.

Suggested commands (macOS / Linux):

```
python -m venv .venv
source .venv/bin/activate
pip install flask pandas
python app.py
```

Report update note: this document is based on the uploaded prior report and the repository/HTML references accessible during preparation. The GitHub HTML file is approximately 8.68 MB and its full source was not available in the rendered web preview, so specific interface changes inside that file have not been asserted without evidence.